

After completing the preparation materials, students should be able to:

1. Interpret the pharmacokinetic properties of the antihypertensive agents in the context of their clinical significance.
2. Relate the mechanisms of action of the ACE inhibitors, AT1 receptor blockers, and direct renin inhibitors to their effects on vascular tone, renal function, and cardiovascular structure by giving examples of therapeutic applications.
3. Compare and contrast the adverse effects, drug interactions, and contraindications when given the mechanism of action and/or pharmacokinetic properties of the individual drug classes.
4. Select the appropriate antihypertensive agent and monitoring parameters for the individual patient based on the drug's pharmacologic effects when presented with a clinical case vignette.

What you need to learn about:

Prerequisites: To understand the mechanisms and effects of therapeutic interventions with blood pressure lowering drugs (the pharmacology of antihypertensive agents), students should have an understanding of normal cardiovascular functions: hemodynamics, systolic and diastolic blood pressure, baroreceptor and humoral regulation of blood pressure, autonomic nervous system response, and the renin-angiotensin-aldosterone system.

- The recommended blood pressure goals
- The antihypertensive agents preferred for initial therapy: calcium channel blockers, angiotensin converting enzyme inhibitors (ACEIs), angiotensin AT-1 receptor blockers (ARBs), and thiazide and thiazide-like diuretics – and why they are preferred initial drugs for the management of primary hypertension and their relative efficacy
- What antihypertensive agents are considered “second-line” and why they are used as adjuncts to the initial drugs

What you need to know

- The mechanism of action of each of the classes of antihypertensive agents
- How each of the “starred” drugs reflects the class characteristics and what properties make them different from the rest of the drugs in the class
- The reflex (compensatory) baroreceptor and humoral responses to the initial and chronic therapy with specific classes and doses of antihypertensive agents
- The patient factors and the potential drug-disease interactions that should be considered when selecting antihypertensive therapy for the individual patient
- The clinically relevant pharmacokinetic properties:
 - By what route will the patient take the medication?
 - How often will the patient have to take it?
 - Can it be given intravenously for the treatment of hypertensive emergencies?
 - What are the potential drug-drug interactions and their mechanisms?
- The additional non-hypertension uses of each class of drugs
- What to tell the patient when counseling them about their medications

Background:

Regulation of Blood Pressure

All antihypertensive drugs produce effects by interfering with normal mechanisms of BP regulation:

- Resistance arterioles
 - Capacitance venules,
 - Heart (cardiac output),
 - Kidneys (volume)
- Antihypertensive therapy can induce compensatory responses to a drop in blood pressure:
- Increased sympathetic outflow
 - Increased renin secretion

Treatment of Hypertension: Goals

Lifestyle modifications

Drug therapy

Individualize Antihypertensive Therapy

2025 ACC/AHA High Blood Pressure Guidelines

Screening

CVD risk factors in adults with hypertension:

- smoking,
- excessive weight,
- diabetes,
- dyslipidemia,
- low fitness, unhealthy diet, psychosocial stress, sleep apnea

HTN evaluation

- target organ damage
- established CV or renal disease
- CV risk factors
- lifestyle contributors to HTN
- drugs:
sympathomimetics, NSAIDs, drugs containing Na⁺, erythropoietin, angiogenesis inhibitors

Question: Integrating osteopathic principles and practice and OMT in patient care, including pharmacotherapy.

- A 42-year-old person presents with new onset hypertension. What areas might you examine for viscerosomatic reflexes to help you decide if this patient's issue is primary or secondary?

Rational treatment is based upon an understanding of the basic principles of body unity, self-regulation, and the interrelationship of structure and function.

Classes of Antihypertensive Agents

Antihypertensive Agents Video Lectures		Diuretics (see lecture on diuretics pharmacology)
	Part 1	Angiotensin converting enzyme inhibitors (ACEIs) Angiotensin II AT1 receptor antagonists (ARBs) Renin inhibitors
	Part 2	Calcium channel blockers Beta blockers
	Part 3	Alpha blockers Sympathoplegics (centrally acting) Sympathetic nerve terminal blockers Vasodilators (other)

Antihypertensive Drugs Actions and Classes

Preferred initial therapy	Second-line therapy options
– Angiotensin converting enzyme inhibitors – Angiotensin II AT₁ receptor blockers – Calcium channel blockers (CCB), long-acting – Thiazide-type / thiazide-like diuretics	– Beta blockers – Alpha blockers – Alpha-2 agonists – Vasodilators – Direct renin inhibitor

Initial Antihypertensive Therapy General Recommendations

The attained blood pressure is the major outcome determinant of reduction in CV risk – not the choice of drug. Therapy should be patient-specific.

Monotherapy: Stage 1 HTN 130-139/80-89 mmHg	Combination therapy: BP >20/10mm Hg above goal or Stage 2 HTN or SBP $\geq$150, DBP $\geq$90 mmHg (Experts differ in their approaches.)
• ACEI or ARB or CCB • Thiazide / thiazide-like diuretic for patients with edema, osteoporosis, calcium nephrolithiasis	Recommendation: • ACEI or ARB + CCB (ACCOMPLISH trial) • ACEI or ARB + thiazide/thiazide-like • CCB + thiazide/thiazide-like diuretic

For patients who require initial combination therapy with two drugs, they should be preferred agents with complementary mechanisms of action, low incidence of side effects, and long-acting, unless the individual patient has an indication for a drug from a different class.

Pharmacokinetics: General Class Properties

ACE inhibitors

ARBs

A	Oral prodrugs, generally moderate absorption	Oral, Good absorption
	Enalaprilat is IV Active compounds have poor absorption.	
D	Moderate to extensive protein binding	Highly bound to plasma proteins
M	Hepatic esterases to active metabolite	Minimal hepatic metabolism
	(except captopril and lisinopril - not prodrugs) Some are further metabolized.	Some are minor substrates of CYP2C9 Some have active metabolite
F	Bioavailability varies	Bioavailability range: 10-80% Irbesartan 60-80%; the rest have <50%
E	Urine and/or feces	Feces and urine
	Some ACEIs require renal dosing adjustment	Some ARBs require renal dosing adjustment
t _{1/2}	Serum half-life varies	Range ~6 hours-12 h (adults) Longer in elderly
	Peak concentrations vary widely, 1-10 hours Captopril 2 to 3 doses per day. All others may be given once daily for HTN mgmt	Onset: 1-2 hours 1x daily dosing for HTN mgmt

ACE inhibitors

Competitive inhibitors of angiotensin converting enzyme

- Inhibits conversion of ANG I to ANG II
- Reduces ANG II effects at both AT1 and AT2 receptors
- Inhibits bradykinin and substance P degradation (vasodilating peptide / neuropeptide)

ARBs

Competitive antagonists of angiotensin II AT1 receptor inhibitors

- High-affinity binding
- Permit activation of AT2 receptors
- Stimulate renin release, which increases circulating ANG II levels, but ARBs block ANG II effects at AT1 receptors

Effects of both ACEIs and ARBs:

- ↓ arterial and venous pressure and peripheral vascular resistance; reduces preload, afterload
- heart rate and cardiac output change very little
may cause reset of baroreceptor reflex to lower arterial pressure
- ↑ natriuresis by increasing renal blood flow and reducing aldosterone and ADH secretion
- ↓ resistance in efferent glomerular arteriole → ↓ glomerular capillary pressure and ↓ proteinuria
- prevent remodeling of smooth muscle and cardiac myocytes

ACEIs and ARBs provide similar benefits in patients with hypertension, heart failure, and chronic kidney disease.

Efficacy is enhanced by concomitant diuretic therapy and reduced dietary sodium intake (as with other RAS inhibitors).

ACEIs / ARBs: Use in Hypertension

ACEI or ARB are included in Guidelines as a preferred initial therapy option

Alone or in combination with other antihypertensive agent(s)

Start with low dose

Potential for a large initial drop in blood pressure (hypotension), especially in patients who may have an active RAS

(e.g. diuretic-induced volume contraction or congestive heart failure)

Efficacy: Lower blood pressure to some extent in most patients with hypertension

Full effect usually occurs after several weeks

Left ventricular hypertrophy regression is reported with ACEIs, ARBs.

ACEIs / ARBs: Additional Therapeutic Uses

ACEIs may have benefits in addition to lowering the blood pressure in:

Post-myocardial infarction	Beneficial effects on mortality after acute MI ACEIs induce sustained BP lowering, regression of ventricular hypertrophy, and prevent ventricular remodeling after myocardial infarction; other mechanisms may contribute.
Heart failure with reduced ejection fraction	Improve symptoms, reduce hospitalizations, and prolong survival • With beta blocker (loop diuretic for fluid control in overt HF) ACEI, ARB, or neprilysin inhibitor-ARB (secubitril-valsartan)
Proteinuric chronic kidney disease, diabetic and nondiabetic	Slow progression of disease Reduce glomerular pressure by reducing BP and cause dilation of efferent arterioles Reduce protein excretion

ACEIs / ARBs: Adverse effects

Effect	Mechanism
Hypotension	Usually occurs with first few doses Most often observed in a RAAS-dependent state, such as volume-depleted patients
Reduced GFR	Relaxed efferent arteriole Usually modest; may be severe in patients whose intrarenal perfusion pressure is already reduced.
Hyperkalemia	ACEIs and ARBs reduce aldosterone secretion, which impairs K ⁺ excretion in the urine. (ANGII, also plasma K ⁺ elevation, stimulate aldosterone secretion.

ACEIs / ARBs: Adverse effects

Effect	Mechanism
Dry cough	ACEIs: 5% to 20% of patients accumulation in lungs of bradykinin, substance P or prostaglandins Low incidence with ARB therapy.
Angioedema	ACEIs: Rare (0.1% to 0.7%), potentially fatal Bradykinin and substance P seem to be involved (inflammatory vasoactive peptides → vasodilation) ARBs: Incidence rates are not higher than with non-RAS antihypertensive drugs.
Enteropathy, sprue-like	Olmesartan characterized by severe chronic diarrhea and weight loss

ACEI / ARB Drug-Drug Interactions

- ACEI + ARB
- ACE inhibitor or ARB + direct renin inhibitor

→ Increased adverse effects

**Potentially life-threatening
hypotension, syncope, hyperkalemia, kidney dysfunction**

Concomitant RAS inhibitor therapy is
NOT RECOMMENDED

- K⁺-sparing drugs: ENaC inhibitors, MR antagonists, NSAIDs
- NSAIDs also blunt the antihypertensive effects by increasing reabsorption of Na⁺ and water.
- Lithium levels may increase requiring lower lithium dose (monitor).

Aliskiren Properties

Chem	Small non-peptide drug	
PK	• Oral • P-glycoprotein substrate; bioavailability ~3% • Extent of metabolism unknown, may involve CYP3A4 • Excreted mainly unchanged in feces via biliary system • $t_{1/2}$ ~24h / 1x daily dosing	CrCl <30 mL/min: Risk of hyperkalemia is increased. Progressive renal impairment may occur. AUC is increased in patients $\geq$ 65 years.
MOA and effects	Highly potent, selective competitive inhibitor of renin • Dose-dependent decreases in ANG I, ANG II, aldosterone • Decreases BP: Efficacy comparable to other antihypertensive agents • <i>Adverse effects</i> as for other RAS inhibitors: hypotension, hyperkalemia, nephrotoxicity, cough, angioedema, pregnancy contraindication	
Use	Hypertension (alternative for HTN treatment if ACEI or ARB contraindicated or not tolerated) • Monotherapy or in combination with other antihypertensive agents but not combined with ACE inhibitors or ARBs	

Aliskiren Drug Interactions

Potentially significant interactions may exist, requiring dose or frequency adjustment, additional monitoring, and/or selection of alternative therapy.

- Other antihypertensive agents (beneficial therapeutic use)
- **Strong P-glycoprotein inducers or inhibitors** decrease / increase aliskiren systemic levels

➤ ***Itraconazole, cyclosporine, ritonavir should be avoided.***

Combination with ACEI or ARB ***not recommended***:

1. Increased risk of hyperkalemia / hypotension / renal impairment
2. Does not lower the risk of cardiovascular event

Question: You may want to pause the video.

- An 58-year-old non-diabetic person presenting with crushing chest pain is ultimately diagnosed with acute ST-elevation myocardial infarction (STEMI) with anterior infarct. Percutaneous coronary intervention is performed. Post-procedure oral pharmacotherapy initiated prior to discharge consists of low-dose aspirin, clopidogrel, metoprolol succinate, rosuvastatin, and low-dose lisinopril. Blood pressure is carefully monitored.
- The patient is at high risk for what RAS-inhibitor-related adverse effect?

Question: You may want to pause the video.

- The randomized double-blind Aliskiren Trial in Type 2 Diabetes Using Cardiorenal Endpoints (ALTITUDE) assigned 8,561 patients to either aliskiren or placebo as an adjunct to ACE inhibitor or ARB. The primary endpoint was a composite of the time to cardiovascular death or a first occurrence of cardiac arrest with resuscitation; nonfatal myocardial infarction; nonfatal stroke; unplanned hospitalization for heart failure; end-stage renal disease, death attributable to kidney failure, or need for renal replacement therapy with no dialysis or transplantation available or initiated; or doubling of baseline serum creatinine.
- The trial was stopped early after the second interim analysis showing that the addition of aliskiren to standard therapy with RAS blockade in patients with type 2 diabetes at high risk for cardiovascular and renal events is not supported by the data (no benefit in primary endpoints) and might even be harmful.
- Thinking about the mechanism-related adverse effects of RAS inhibition, particularly when RAS inhibitors are combined, what were two toxicities occurring at a significantly higher rate in the aliskiren group compared to the placebo group?

- RAS inhibitors reduce peripheral vascular resistance without reflex sympathetic activation and they decrease aldosterone secretion.
- ACEIs increase vasodilating peptides bradykinin and substance P levels, which may also be associated with causing cough and angioedema. ARBs are associated with a low incidence of cough and angioedema. DRIs can cause these adverse effects.
- ACE inhibitors and ARBs are effective and widely used in the treatment of hypertension, post-acute myocardial infarction therapy, heart failure with reduced ejection fraction, and proteinuric chronic kidney disease. Aliskiren is an option for the management of hypertension. It is not a recommended first-line therapy.
- RAS inhibitors are generally well tolerated.
- RAS inhibitors can cause hypotension; patients with RAAS activation are at high risk. Other adverse effects include modest reduction in GFR, hyperkalemia, and nephrotoxicity. ACEIs are associated with a dry cough and angioedema that may be related to elevated levels of bradykinin and substance P.
- Olmesartan is associated with sprue-like enteropathy with severe diarrhea and weight loss.
- **Combinations of RAS inhibitors should be avoided due to increased risk of hypotension, syncope, renal impairment, and hyperkalemia.**
- RAS inhibitors are contraindicated during pregnancy because of the risk of fetal hypotension, anuria and renal failure leading to oligohydramnios. Fetal malformations and death may occur.